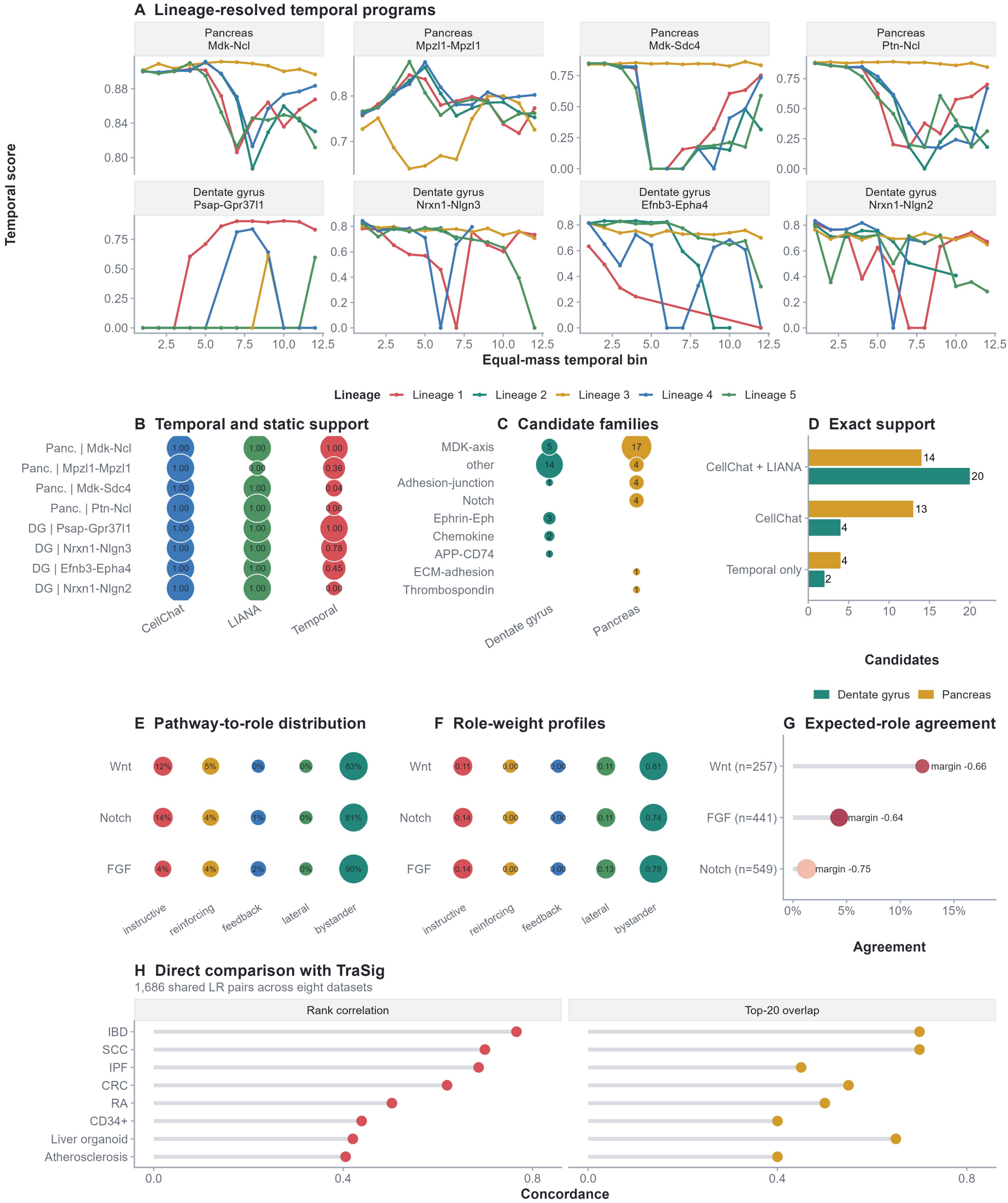

A Lineage-resolved temporal programs
Pancreas
Mdk-Ncl
Pancreas
Mpzl1-Mpzl1
Pancreas
Mdk-Sdc4
Pancreas
Ptn-Ncl
Dentate gyrus
Psap-Gpr37l1
Dentate gyrus
Nrxn1-Nlgn3
Dentate gyrus
Efnb3-Epha4
Dentate gyrus
Nrxn1-Nlgn2
Temporal score
Equal-mass temporal bin
Lineage
Lineage 1
Lineage 2
Lineage 3
Lineage 4
Lineage 5
B Temporal and static support
Panc. | Mdk-Ncl
Panc. | Mpzl1-Mpzl1
Panc. | Mdk-Sdc4
Panc. | Ptn-Ncl
DG | Psap-Gpr37l1
DG | Nrxn1-Nlgn3
DG | Efnb3-Epha4
DG | Nrxn1-Nlgn2
CellChat
LIANA
Temporal
C Candidate families
MDK-axis
other
Adhesion-junction
Notch
Ephrin-Eph
Chemokine
APP-CD74
ECM-adhesion
Thrombospondin
Dentate gyrus
Pancreas
D Exact support
CellChat + LIANA
CellChat
Temporal only
Candidates
Dentate gyrus
Pancreas
E Pathway-to-role distribution
Wnt
Notch
FGF
instructive
reinforcing
feedback
lateral
bystander
F Role-weight profiles
G Expected-role agreement
Wnt (n=257)
FGF (n=441)
Notch (n=549)
margin -0.66
margin -0.64
margin -0.75
Agreement
H Direct comparison with TraSig
1,686 shared LR pairs across eight datasets
Rank correlation
Top-20 overlap
IBD
SCC
IPF
CRC
RA
CD34+
Liver organoid
Atherosclerosis
Concordance